

## WandiSSAL\_LoggerAppsMsg()

### Synopsis

```
#include "WandiSSAL-LoggerLib-2.0.h"
```

```
int   WandiSSAL_LoggerAppsMsg(int devProd, int message_group, int severity,  
char *message_string, ... )
```

### Description

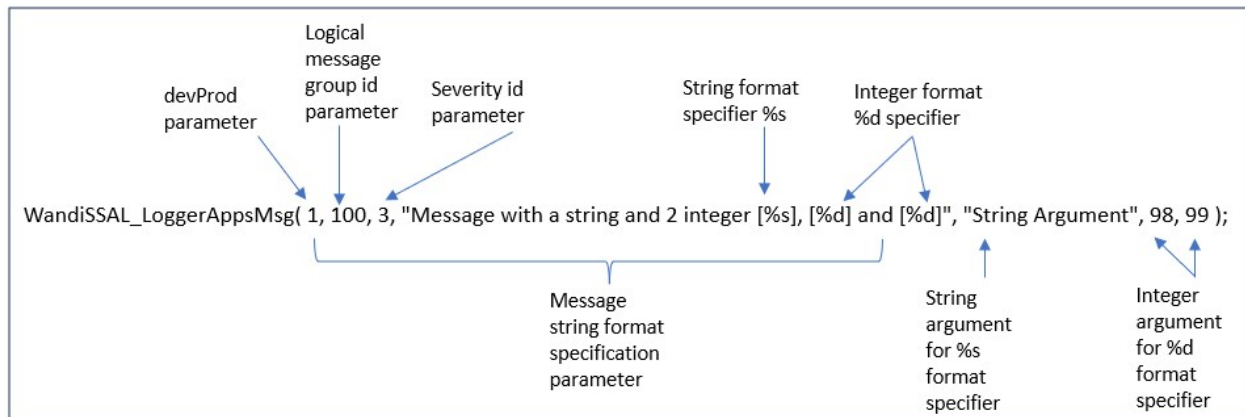
WandiSSAL\_LoggerAppsMsg() decomposes argument parameters into id=value pairs so that it can validate for errors, encode id=value pairs to reduce storage capacity, and securely protect id=value pairs to ensure confidentiality, integrity, and authenticity.<sup>1</sup>

### Example Usage

```
WandiSSAL_LoggerAppsMsg(1,100,3,"Message that generates 1 string and 2  
integers [%s], [%d] and [%d]","String Argument", 98,99);
```

Generates formatted output log message:

```
02/09/2026 12:01:55 [3650:sample_program] [DATABASE] [INFORM] Message that  
generates 1 string and 2 integers [String Argument], [98] and [99]
```



The `devProd` integer parameter 1 indicates a development log message and 2 indicates a production log message. This will enable generating log messages 1 and 2 together or 2 only. The benefit of this is to prevent generating unnecessary development log messages in a production environment.

The `logical message_group` integer parameter is the id part of the id=value pairs defined in file `LoggerMessageDefines.txt`. The id part of 100 is translated and mapped to the corresponding value part of `DATABASE` in the output message. Refer to the `LoggerMessageDefines.txt` document for further details.

The `severity` integer parameter is the id part of the id=value pairs defined in file `LoggerMessageDefines.txt`. The id part of 3 is translated and mapped to the corresponding value

---

<sup>1</sup> This free version only supports decompose and validate format string specification and string argument parameters.

part of INFORM in the output message. Refer to the `LoggerMessageDefines.txt` document for further details.

The `message_string` format string specification parameter is a C-Programming Language `printf()` style format string specification that comprises of format string text and format specifiers. This `message_string` parameter is bounded by the guidelines in the `Wandi-SSAL-Bound-Capabilities.pdf` document.

The reminder of the `WandiSSAL_loggerAppsMsg()` parameters (as indicated by "...") represents the corresponding format string specifiers argument values for `message_string`.

NOTE: Refer to the `Wandi-SSAL-Bound-Capabilities.pdf` document for validation guidelines used by `WandiSSAL_loggerAppsMsg()` to check and validate parameter values.

`WandiSSAL_loggerAppsMsg()` returns 0 on success and -1 on error.

This API function is failed safe which means when an error occurs the application can continue processing.

### **Error Codes**

Error codes are written to the following files, where the `<cpid>` is the application process id number:

- `/tmp/<cpid>-LoggerAppsMessage.txt`
- `$LOGGER_HOME_PATH/LoggerOutput/<cpid>-LoggerAppsMessage.txt`

|               |                                                                                                                                                                                                                 |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 135           | Invalid <code>devProd</code> parameter.<br><b>Solution:</b> Refer to <code>devProd</code> guidelines above.                                                                                                     |
| 136           | Invalid <code>message_group</code> , <code>severity</code> and <code>message_string</code> parameter.<br><b>Solution:</b> Refer to <code>message_group</code> and <code>message_string</code> guidelines above. |
| 137           | Invalid <code>message_string</code> parameter.<br><b>Solution:</b> Refer to <code>message_string</code> guidelines above                                                                                        |
| 138           | Too many format string type specifiers.<br><b>Solution:</b> Refer to <code>Wandi-SSAL-Bound-Capabilities.pdf</code> guidelines.                                                                                 |
| 139           | String argument pointer parameter is NULL.<br><b>Solution:</b> Make sure string argument pointer is not NULL.                                                                                                   |
| 140           | String argument pointer is -1.<br><b>Solution:</b> Make sure string argument pointer is not -1.                                                                                                                 |
| 141           | String argument not properly NULL terminated.<br><b>Solution:</b> Refer to <code>Wandi-SSAL-Bound-Capabilities.pdf</code> guidelines.                                                                           |
| 142           | String argument exceeded allowed length of 64 characters.<br><b>Solution:</b> Refer to <code>Wandi-SSAL-Bound-Capabilities.pdf</code> guidelines.                                                               |
| 143           | Invalid string argument value.<br><b>Solution:</b> Refer to <code>Wandi-SSAL-Bound-Capabilities.pdf</code> guidelines.                                                                                          |
| 144           | Invalid format string specifier type.<br><b>Solution:</b> Refer to <code>Wandi-SSAL-Bound-Capabilities.pdf</code> guidelines.                                                                                   |
| 115, 117      | Parameter <code>message_string</code> is NULL or exceeded allowed length of 256 characters.<br><b>Solution:</b> Refer to <code>Wandi-SSAL-Bound-Capabilities.pdf</code> guidelines.                             |
| 104, 106, 116 | Invalid ASCII character in <code>message_string</code> or string argument.                                                                                                                                      |

|     |                                                                                                   |
|-----|---------------------------------------------------------------------------------------------------|
|     | <b>Solution:</b> Refer to <code>Wandi-SSAL-Bound-Capabilities.pdf</code> guidelines.              |
| 105 | String argument length is zero.<br><b>Solution:</b> Make sure string argument is not zero length. |